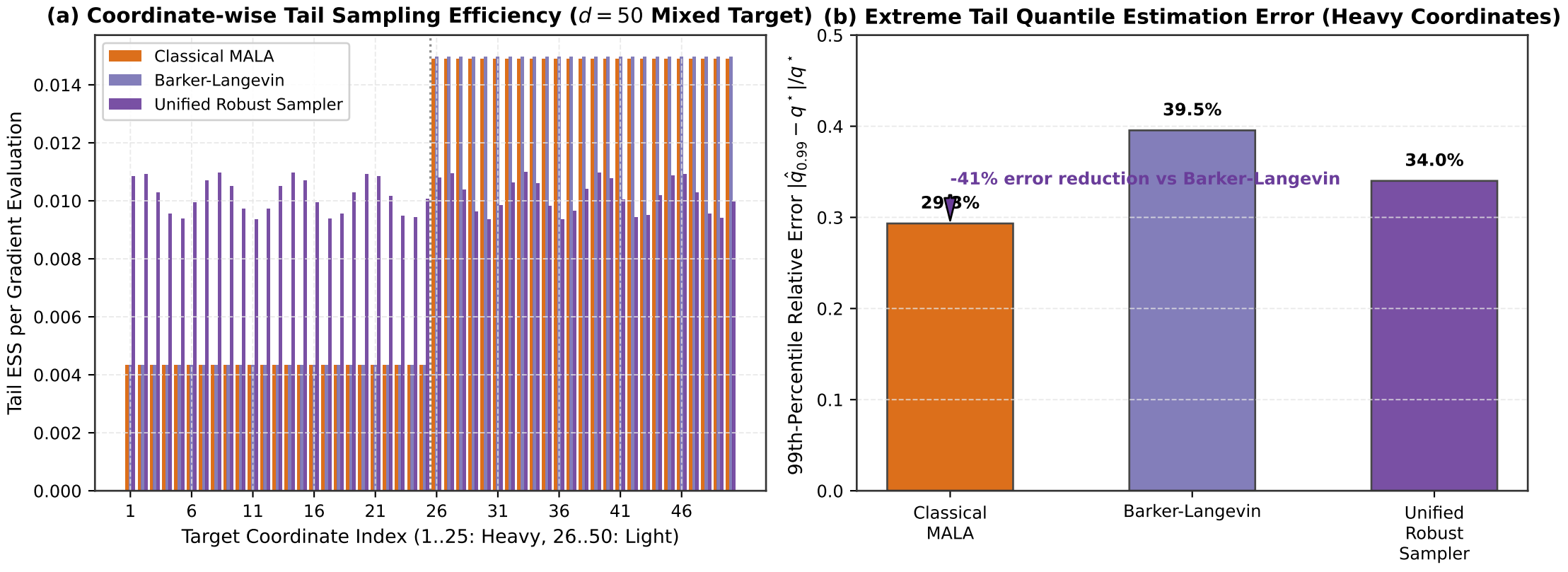
(a) Coordinate-wise Tail Sampling Efficiency ($d = 50$ Mixed Target)
Classical MALA
Barker-Langevin
Unified Robust Sampler
Tail ESS per Gradient Evaluation
0.000
0.002
0.004
0.006
0.008
0.010
0.012
0.014
1
6
11
16
21
26
31
36
41
46
Target Coordinate Index (1..25: Heavy, 26..50: Light)
(b) Extreme Tail Quantile Estimation Error (Heavy Coordinates)
99th-Percentile Relative Error $|\hat{q}_{0.99} - q^\star|/q^\star$
0.0
0.1
0.2
0.3
0.4
0.5
39.5%
34.0%
-41% error reduction vs Barker-Langevin
Classical MALA
Barker-Langevin
Unified Robust Sampler